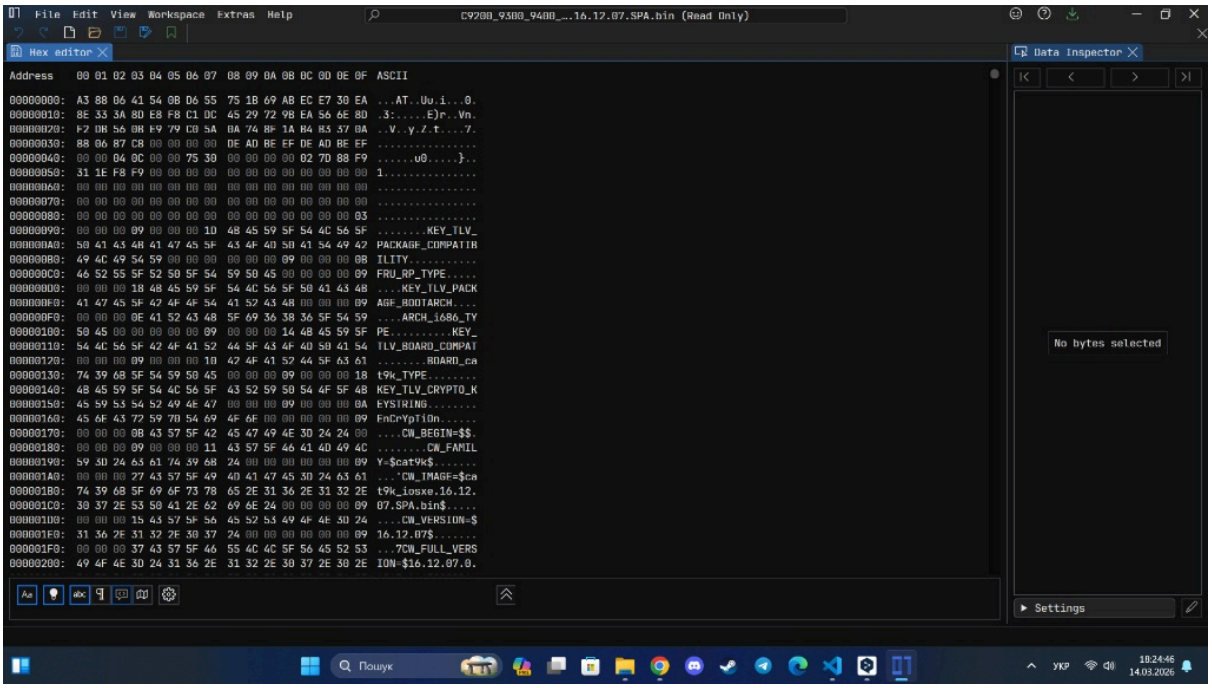
Before starting the task, I looked for some general information about Go. I watched a few videos on the tylerlaceby channel about writing a parser.

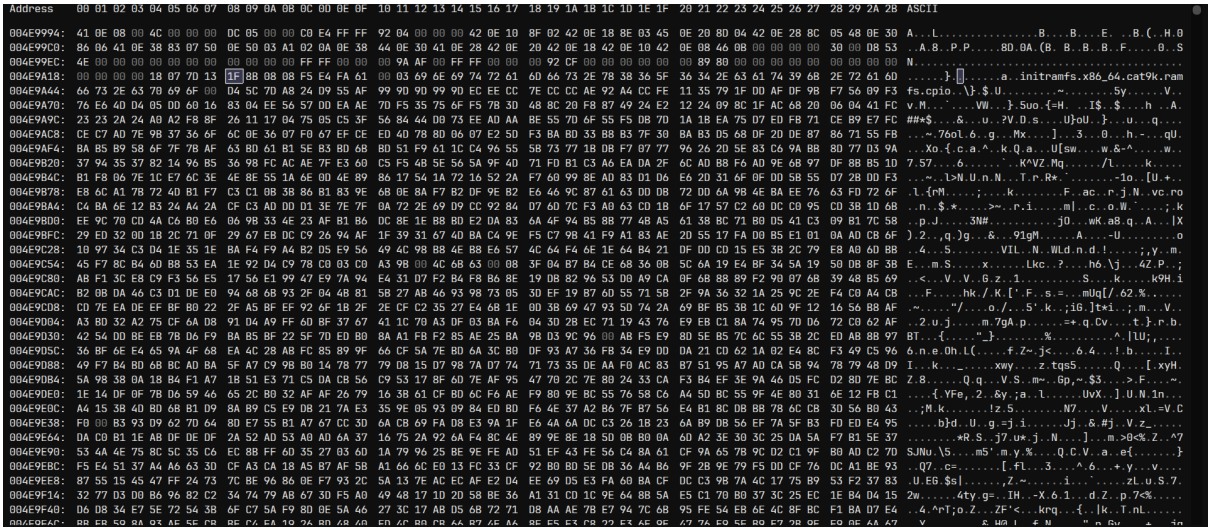
Later, I used Gemini to search for information, correct the code from the videos, and get a general plan for completing the task. From it, I learnt about the ImHex programme, which I used for an initial analysis of the file C9200_9300_9400_9500_9600_cat9k_iosxe.16.12.07.SPA.bin.



Gemini indicated that the first 800 bytes contain information about Cisco (Header), followed by the Linux bzImage kernel, and after the kernel lies the SquashFS file system. To locate the kernel and precisely access SquashFS, I mathematically calculated the number of bytes required.

The bzImage kernel consists of two parts: setup and payload. To understand how to calculate the bzImage, I opened "The Linux/x86 Boot Protocol", where I found their sizes, namely offsets 0x01F1 and 0x01F4. Up to 0x0000080F is the Header; the kernel starts at 0x00000810. I added 0x0810 + 0x01F1 = 0xA01 for the setup size, and 0x0810 + 0x01F4 = 0xA04 for the payload size. Using Gemini, I converted the resulting numbers to bytes and obtained 15,872 (0x3E00) and

5,133,328 (0x4E5410). In the end, I got 0x4E9210. I added 0x0810 to the result and got 0x004E9A20. Looking at it in ImHex, I saw initramfs. So this is where the Linux kernel ends and SquashFS begins.



```
Address   00 01 02 03 04 05 06 07  08 09 0A 0B 0C 0D 0E 0F  10 11 12 13 14 15 16 17  18 19 1A 1B 1C 1D 1E 1F  20 21 22 23 24 25 26 27  28 29 2A 2B  ASCII

004E9994: 41 0E 08 00 4C 00 00 00  DC 05 00 00 C0 E4 FF FF  92 04 00 00 00 42 0E 10  8F 02 42 0E 18 8E 03 45  0E 20 8D 04 42 0E 28 8C  05 48 0E 30  A...L...............B....B....E. ..B.(..H.0
004E99C0: 86 06 41 0E 38 83 07 50  0E 50 03 A1 02 0A 0E 38  44 0E 30 41 0E 28 42 0E  20 42 0E 18 42 0E 10 42  0E 08 46 0B 00 00 00 00  30 00 D8 53  ..A.8..P.P.....8D.0A.(B. B..B..B..F.....0..S
004E99EC: 4E 00 00 00 00 00 00 00  00 00 00 00 FF FF 00 00  00 9A AF 00 FF FF 00 00  00 92 CF 00 00 00 00 00  00 89 B0 00 00 00 00 00  00 00 00 00  N.......................................
004E9A18: 00 00 00 00 18 07 7D 13  1F 8B 08 08 F5 E4 FA 61  00 03 69 6E 69 74 72 61  6D 66 73 2E 78 38 36 5F  36 34 2E 63 61 74 39 6B  2E 72 61 6D  ......}.[].....a..initramfs.x86_64.cat9k.ram
004E9A44: 66 73 2E 63 70 69 6F 00  D4 5C 7D A8 24 D9 55 AF  99 9D 9D 99 9D EC EE CC  7E CC CC AE 92 A4 CC FE  11 35 79 1F DD AF DF 9B  F7 56 09 F3  fs.cpio..\}.$.U.........~........5y.....V..
004E9A70: 76 E6 4D D4 05 DD 60 16  83 04 EE 56 57 DD EA AE  7D F5 35 75 6F F5 7B 3D  48 8C 20 F8 87 49 24 E2  12 24 09 8C 1F AC 68 20  06 04 41 FC  v.M...`....VW...}.5uo.{=H. ..I$..$....h ..A.
004E9A9C: 23 23 2A 24 A0 A2 F8 8F  26 11 17 04 75 05 C5 3F  56 84 44 D0 73 EE AD AA  BE 55 7D 6F 55 F5 DB 7D  1A 1B EA 75 D7 ED FB 71  CE B9 E7 FC  ##*$....&...u..?V.D.s....U}oU..}...u...q....
004E9AC8: CE C7 AD 7E 9B 37 36 6F  6C 0E 36 07 F0 67 EF CE  ED 4D 78 8D 06 07 E2 5D  F3 BA BD 33 B8 B3 7F 30  BA B3 D5 68 DF 2D DE 87  86 71 55 FB  ...~.76ol.6..g...Mx....]...3...0...h.-...qU.
004E9AF4: BA B5 B9 58 6F 7F 7B AF  63 BD 61 B1 5E B3 BD 6B  BD 51 F9 61 1C C4 96 55  5B 73 77 1B DB F7 07 77  96 26 2D 5E 83 C6 9A BB  8D 77 D3 9A  ...Xo.{.c.a.^..k.Q.a...U[sw....w.&-^.....w..
004E9B20: 37 94 35 37 82 14 96 B5  36 98 FC AC AE 7F E3 60  C5 F5 4B 5E 56 5A 9F 4D  71 FD B1 C3 A6 EA DA 2F  6C AD B8 F6 AD 9E 6B 97  DF 8B B5 1D  7.57....6......`..K^VZ.Mq....../l.....k.....
004E9B4C: B1 F8 06 7E 1C E7 6C 3E  4E 8E 55 1A 6E 0D 4E 89  86 17 54 1A 72 16 52 2A  F7 60 99 8E AD 83 D1 D6  E6 2D 31 6F 0F DD 5B 55  D7 2B DD F3  ...~..l>N.U.n.N...T.r.R*.`.......-1o..[U.+..
004E9B78: E8 6C A1 7B 72 4D B1 F7  C3 C1 0B 3B 86 B1 83 9E  6B 0E 8A F7 B2 DF 9E B2  E6 46 9C 87 61 63 DD DB  72 DD 6A 9B 4E BA EE 76  63 FD 72 6F  .l.{rM.....;....k........F..ac..r.j.N..vc.ro
004E9BA4: C4 BA 6E 12 B3 24 A4 2A  CF C3 AD DD D1 3E 7E 7F  0A 72 2E 69 D9 CC 92 84  D7 6D 7C F3 A0 63 CD 1B  6F 17 57 C2 60 DC C0 95  CD 3B 1D 6B  ..n..$.*.....>~..r.i.....m|..c..o.W.`....;.k
004E9BD0: EE 9C 70 CD 4A C6 B0 E6  06 9B 33 4E 23 AF B1 B6  DC 8E 1E B8 BD E2 DA 83  6A 4F 94 B5 8B 77 48 A5  61 38 BC 71 B0 D5 41 C3  09 B1 7C 58  ..p.J.....3N#..........jO...wH.a8.q..A...|X
004E9BFC: 29 ED 32 0D 1B 2C 71 0F  29 67 EB DC C9 26 94 AF  1F 39 31 67 4D BA C4 9E  F5 C7 9B 41 F9 A1 83 AE  2D 55 17 FA D0 B5 E1 01  0A AD C8 6F  ).2..,q.)g...&...91gM.....A....-U.........o
004E9C28: 10 97 34 C3 D4 1E 35 1E  BA F4 F9 A4 B2 D5 E9 56  49 4C 98 B8 4E B8 E6 57  4C 64 F4 6E 1E 64 84 21  DF DD CD 15 E5 3B 2C 79  E8 A0 6D BB  ..4...5........VIL..N..WLd.n.d.!.....;,y..m.
004E9C54: 45 F7 8C B4 6D B8 53 EA  1E 92 D4 C9 78 C0 03 C0  A3 9B 00 4C 6B 63 00 08  3F 04 B7 B4 CE 68 36 0B  5C 6A 19 E4 BF 34 5A 19  50 DB 8F 3B  E...m.S.....x......Lkc..?....h6.\j...4Z.P..;
004E9C80: AB F1 3C E8 C9 F3 56 E5  17 56 E1 99 47 E9 7A 94  E4 31 D7 F2 B4 F8 B6 8E  19 DB 82 96 53 D0 A9 CA  0F 6B 88 89 F2 90 07 6B  39 48 B5 69  ..<...V..V..G.z..1.........S....k.....k9H.i
004E9CAC: B2 0B DA 46 C3 D1 DE E0  94 68 6B 93 2F 04 4B 81  5B 27 AB 46 93 98 73 05  3D EF 19 B7 6D 55 71 5B  2F 9A 36 32 1A 25 9C 2E  F4 C0 A4 CB  ...F.....hk./.K.['.F..s.=...mUq[/.62.%.....
004E9CD8: CD 7E EA DE EF BF B0 22  2F A5 BF EF 92 6F 1B 2F  2E CF C2 35 27 E4 6B 1E  0D 3B 69 47 93 5D 74 2A  69 BF B5 3B 1C 6D 9F 12  16 56 B8 AF  .~.....“/....o./...5'.k..;iG.]t*i..;.m...V..
004E9D04: A3 BD 32 A2 75 CF 6A D8  91 D4 A9 FF 6D BF 37 67  41 1C 70 A3 DF 03 BA F6  04 3D 2B EC 71 19 43 76  E9 EB C1 8A 74 95 7D D6  72 C0 62 AF  ..2.u.j.....m.7gA.p.....=+.q.Cv...t.}.r.b.
004E9D30: 42 54 DD BE EB 7B D6 F9  BA B5 BF 22 5F 7D ED B0  8A A1 FB F2 85 AE 25 BA  9B D3 9C 96 00 AB F5 E9  8D 5E B5 7C 6C 55 3B 2C  ED AB 88 97  BT...{.....“_}.......%.........^.|lU;,....
004E9D5C: 36 BF 6E E4 65 9A 4F 68  EA 4C 28 AB FC 85 89 9F  66 CF 5A 7E BD 6A 3C B0  DF 93 A7 36 FB 34 E9 DD  DA 21 CD 62 1A 02 E4 8C  F3 49 C5 96  6.n.e.Oh.L(.....f.Z~.j<....6.4...!.b.....I..
004E9D88: 49 F7 B4 BD 6B BC AD BA  5F A7 C9 9B B0 14 78 77  79 D8 15 D7 98 7A D7 74  71 73 35 DE AA F0 AC 83  B7 51 95 A7 AD CA 5B 94  78 79 48 D9  I..k..._.....xwy....z.tqs5.....Q....[.xyH.
004E9D84: 5A 98 38 0A 18 B4 F1 A7  1B 51 E3 71 C5 DA CB 56  C9 53 17 8F 6D 7E AF 95  47 70 2C 7E 80 24 33 CA  F3 B4 FF 3E 9A 46 D5 FC  D2 8D 7E BC  Z.8.....Q.q..V.S..m~.Gp,~.$3...>.F....~..
004E9DE0: 1E 14 DF 0F 7B D6 59 46  65 2C B0 32 AF AF 26 79  16 3B 61 CF BD 6C F6 AE  F9 80 9E BC 55 76 58 C6  A4 5D BC 55 9F 4E 80 31  6E 12 FB C1  ...{.YFe,.2..&y.;a..l....UvX..].U.N.1n...
004E9E0C: A4 15 3B 4D BD 6B B1 D9  8A B9 C5 E9 DB 21 7A E3  35 9E 05 93 09 84 ED BD  F6 4E 37 A2 B6 7F B7 56  E4 B1 8C DB BB 78 6C CB  3D 56 B0 43  ..;M.k.....!z.5.........N7...V....xl.=V.C
004E9E38: F0 00 B3 93 D9 62 7D 64  8D E7 55 B1 A7 67 CC 3D  6A CB 69 FA D8 E3 9A 1F  E6 4A 6A DC C3 26 1B 23  6A B9 DB 56 EF 7A 5F B3  FD ED E4 95  .....b}d..U..g.=j.i.....Jj..&.#j..V.z_....
004E9E64: DA C0 B1 1E AB DF DE DF  2A 52 AD 53 A0 A0 6A 37  16 75 2A 92 6A F4 8C 4E  89 9E 8E 18 5D 0B B0 0A  6D A2 3E 30 3C 25 DA 5A  F7 B1 5E 37  ........*R.S..j7.u*.j..N....]...m.>0<%.Z..^7
004E9E90: 53 4A 4E 75 8C 5C 35 C6  EC 88 FF 6D 35 27 03 6D  1A 79 96 25 BE 9E FE AD  51 EF 43 FE 56 C4 8A 61  CF 9A 65 7B 9C D2 C1 9F  B0 AD C2 7D  SJNu.\5....m5'.m.y.%....Q.C.V..a..e{.......}
004E9EBC: F5 E4 51 37 A4 A6 63 3D  CF A3 CA 18 A5 B7 AF 5B  A1 66 6C E0 13 FC 33 CF  92 B0 BD 5E DB 36 A4 B6  9F 2B 9E 79 F5 DD CF 76  DC A1 BE 93  ..Q7..c=.......[.fl...3....^.6...+.y...v....
004E9EE8: 87 55 15 45 47 FF 24 73  7C BE 96 86 0E F7 93 2C  5A 13 7E AC EC AF E2 D4  EE 69 D5 E3 FA 60 BA CF  DC C3 9B 7A 4C 17 75 B9  53 F2 37 83  .U.EG.$s|......,Z.~......i...`.....zL.u.S.7.
004E9F14: 32 77 D3 D0 B6 96 82 C2  34 74 79 AB 67 3D F5 A0  49 48 17 1D 2D 58 BE 36  A1 31 CD 1C 9E 64 88 5A  E5 C1 70 B0 37 3C 25 EC  1E B4 D4 15  2w......4ty.g=..IH..-X.6.1...d.Z..p.7<%.....
004E9F40: D6 D8 34 E7 5E 72 54 38  6F C7 5A F9 8D 0E 5A 46  27 3C 17 AB D5 6B 72 71  D8 AA AE 7B E7 94 7C 6B  95 FE 54 EB 6E 4C 8F BC  F1 8A D7 E4  ..4.^rT;o.Z...ZF'<...krq...{..|k..T.nL......
004E9F6C: BB EB 59 8A 93 AF 5E CB  BE C6 EA 19 7A B0 68 40  FD 4C B0 CB A6 B7 4E A6  BE E5 E3 C8 22 E3 AE 9E  47 7A E9 5E B0 E7 2B 9E  E9 0E AA A7  ..Y.....^.....z.h@.L.....N.....".....Gz.^..+.....
```

From that point on, I couldn’t figure out the right way to do it, so I decided to copy the data to a separate file called 2_filesystem.squashfs.